

Airbnb

You are a Data Scientist on the Airbnb Stays team, focusing on analyzing guest booking behaviors....

Question 1: What is the average booking completion rate for properties with 'low' and 'high' pricing transparency levels during April 2024? Use a 7-day rolling window to capture short-term trends. This analysis will help us understand the impact of pricing transparency on booking conversions.

```
WITH calculate AS (  
  SELECT  
    pricing_transparency_level,  
    booking_date,  
    SUM(CASE WHEN completion_status = 'completed' THEN 1 ELSE  
0 END) * 1.0 / COUNT(*) AS daily_completion_rate  
  FROM fct_bookings  
  WHERE TO_CHAR(booking_date, 'YYYY-MM') = '2024-04'  
    AND pricing_transparency_level IN ('low', 'high')  
  GROUP BY pricing_transparency_level, booking_date  
)  
  
SELECT  
  pricing_transparency_level,  
  booking_date,  
  ROUND(AVG(daily_completion_rate) OVER (  
    PARTITION BY pricing_transparency_level  
    ORDER BY booking_date  
    ROWS BETWEEN 6 PRECEDING AND CURRENT ROW  
,3) AS rolling_avg_completion_rate  
FROM calculate  
ORDER BY pricing_transparency_level, booking_date
```



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Question 2: What are the day-over-day changes number of bookings for properties with varying cancellation policies in April 2024? Compare each day with the previous day to reveal trends over time. This analysis will help us assess how cancellation policies influence booking volume on a daily basis.

```
with calculate AS(
  SELECT
    booking_date,
    cancellation_policy,
    COUNT(*) AS number_bookings,
    COALESCE(LAG(COUNT(*)) OVER (
      PARTITION BY cancellation_policy
      ORDER BY booking_date
    ),0) AS prev_day_bookings
  FROM fct_bookings
  WHERE TO_CHAR(booking_date, 'YYYY-MM') = '2024-04'
  GROUP BY booking_date, cancellation_policy
  ORDER BY cancellation_policy, booking_date)

SELECT booking_date, cancellation_policy,
       COALESCE((number_bookings - prev_day_bookings), number_bookings) AS dod_change
FROM calculate
```



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Question 3: What is the percentage difference in booking completion rates between properties with high pricing transparency and those with low transparency in April 2024, when also accounting for differing cancellation policies?

Compute the booking completion rates separately for high and low transparency levels within each cancellation policy, then calculate the percentage difference between them to compare the results.

```
with calculate AS(
  SELECT
    pricing_transparency_level,
    cancellation_policy,
    SUM(CASE WHEN completion_status = 'completed' THEN 1 ELSE
0 END) * 100.0 / COUNT(*) AS completion_rate
  FROM fct_bookings
  WHERE TO_CHAR(booking_date, 'YYYY-MM') = '2024-04'
    AND pricing_transparency_level IN ('low','high')
  GROUP BY pricing_transparency_level, cancellation_policy
  ORDER BY cancellation_policy, pricing_transparency_level),

change AS(
  SELECT
    cancellation_policy,
    MAX(CASE WHEN pricing_transparency_level = 'high' THEN com
pletion_rate END) AS high_rate,
    MAX(CASE WHEN pricing_transparency_level = 'low' THEN comp
letion_rate END) AS low_rate
  FROM calculate
  GROUP BY cancellation_policy)

SELECT *,
  ((high_rate - low_rate) / NULLIF(low_rate, 0)) * 100.0 AS pe
rcentage_difference
FROM change
```

